FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

Computer System and Architecture

FLAB3

ZACK GABRIEL J. VALENZUELA	Sir. Hadji Tejuco
Name of Student	Name of Professor
10/29/2026	10/29/2026
Data Performed	Date Submitted

PROGRAM DESCRIPTION:

The program first asks the user to enter their student information and displays the details entered, then shows the 11 subjects and accepts the units for each subject. Finally, it calculates the total units, tuition at ₱1,000 per unit, and laboratory fees of ₱2,500 per laboratory subject, then displays the estimated total fees.

COMPILATION:

```
and any other fees not specified in the calculation.
@zjvalenzuela-zzzzz ➜/workspaces/FLAB3/FLAB3 (main) $ nasm -f elf32 FLAB3.asm -o FLAB3.o
@zjvalenzuela-zzzzz ➜/workspaces/FLAB3/FLAB3 (main) $ ld -m elf_i386 -e _start FLAB3.o -o FLAB3
@zjvalenzuela-zzzzz ➜/workspaces/FLAB3/FLAB3 (main) $ ./FLAB3
```

PROCESS INPUT/OUTPUT

```
--- ENTER STUDENT INFORMATION ---
Enrollment Status (e.g., ENROLLED):
ENROLLED
Student Number:
202510947
Full Name:
ZACK GABRIEL J. VALENZUELA
Phone Number:
09667283151
ZIP Code:
1420
College:
COMPUTER STUDIES
Program:
BSCSSE
Year Level:
1
Term (e.g., 1st term):
1st term
School Year (e.g., SY 2026-2027):
2026-2027
Complete Address:
BLK 42-A BELGRADE STREET, VISTA VERDE NORTH LLANO ROAD
```

```
--- SUBJECTS AND UNITS ---
Enter 0 units if you are not taking a subject.
1. CCS0021  - INFORMATION MANAGEMENT (LEC)
   Units for this subject (whole number):
2
2. CCS0021L - INFORMATION MANAGEMENT (LAB)
   Units for this subject (whole number):
1
3. CS0001   - DISCRETE STRUCTURES I
   Units for this subject (whole number):
3
4. CS0003   - COMPUTER SYSTEMS & ARCHITECTURE
   Units for this subject (whole number):
2
5. CS0003L  - COMPUTER SYSTEMS & ARCHITECTURE (LAB)
   Units for this subject (whole number):
1
6. CS0070   - OBJECT ORIENTED PROGRAMMING
   Units for this subject (whole number):
2
7. CS0070L  - OBJECT ORIENTED PROGRAMMING-LAB
   Units for this subject (whole number):
1
8. GED0021  - SPECIALIZED ENGLISH PROGRAM 2
   Units for this subject (whole number):
3
9. GED0055  - MATHEMATICAL ANALYSIS
   Units for this subject (whole number):
3
10. GED0081  - COLLEGE PHYSICS 1 LECTURE
   Units for this subject (whole number):
2
11. GED0081L - COLLEGE PHYSICS 1 LABORATORY
   Units for this subject (whole number):
1
```

FINAL OUTPUT

```
--- SUBJECT UNIT SUMMARY ---
1. CCS0021  - INFORMATION MANAGEMENT (LEC)  Units: 2
2. CCS0021L - INFORMATION MANAGEMENT (LAB)  Units: 1
3. CS0001   - DISCRETE STRUCTURES I  Units: 3
4. CS0003   - COMPUTER SYSTEMS & ARCHITECTURE  Units: 2
5. CS0003L  - COMPUTER SYSTEMS & ARCHITECTURE (LAB)  Units: 1
6. CS0070   - OBJECT ORIENTED PROGRAMMING  Units: 2
7. CS0070L  - OBJECT ORIENTED PROGRAMMING-LAB  Units: 1
8. GED0021  - SPECIALIZED ENGLISH PROGRAM 2  Units: 3
9. GED0055  - MATHEMATICAL ANALYSIS  Units: 3
10. GED0081  - COLLEGE PHYSICS 1 LECTURE  Units: 2
11. GED0081L - COLLEGE PHYSICS 1 LABORATORY  Units: 1

--- TUITION FEE SUMMARY ---
Total units: 21
Unit price: PHP 1000
Tuition (total units x PHP 1,000): PHP 21000
Laboratory subjects taken: 4
Laboratory fee (PHP 2,500 each): PHP 10000
TOTAL ESTIMATED TUITION + LAB FEES: PHP 31000

--- STUDENT INFORMATION SUMMARY ---
Enrollment Status: ENROLLED
Student Number: 202510947
Name: ZACK GABRIEL J. VALENZUELA
Phone: 09667283151
ZIP Code: 1420
College: COMPUTER STUDIES
Program: BSCSSE
Year Level: 1
Term: 1st term
School Year: 2026-2027
Address: BLK 42-A BELGRADE STREET, VISTA VERDE NORTH LLANO ROAD
```

SOURCE CODE:

```
; ZACK GABRIEL J. VALENZUELA
; SOLAR / COR ENROLLMENT AND TUITION CALCULATOR
; NASM 32-bit Assembly for Linux
```

```
BITS 32
GLOBAL _start

%macro PRINT 2
   mov eax, 4
   mov ebx, 1
   mov ecx, %1
   mov edx, %2
   int 0x80
%endmacro

%macro ASK_STRING 4
   PRINT %1, %2
   push dword %4
   push dword %3
   call read_line
   add esp, 8
%endmacro

%macro ASK_INTEGER 3
   PRINT %1, %2
   call read_integer
   mov [%3], eax
%endmacro

SECTION .data

   heading db 10, '============================================', 10
           db '      SOLAR / COR TUITION CALCULATOR', 10
           db '============================================', 10, 10
   heading_len equ $ - heading

   student_heading db 10, '--- ENTER STUDENT INFORMATION ---', 10
   student_heading_len equ $ - student_heading

   p_status db 'Enrollment Status (e.g., ENROLLED):', 10
   p_status_len equ $ - p_status

   p_student_no db 'Student Number:', 10
   p_student_no_len equ $ - p_student_no

   p_name db 'Full Name:', 10
   p_name_len equ $ - p_name
```

```
p_phone db 'Phone Number:', 10
p_phone_len equ $ - p_phone

p_zip db 'ZIP Code:', 10
p_zip_len equ $ - p_zip

p_college db 'College:', 10
p_college_len equ $ - p_college

p_program db 'Program:', 10
p_program_len equ $ - p_program

p_year_level db 'Year Level:', 10
p_year_level_len equ $ - p_year_level

p_term db 'Term (e.g., 1st term):', 10
p_term_len equ $ - p_term

p_school_year db 'School Year (e.g., SY 2026-2027):', 10
p_school_year_len equ $ - p_school_year

p_address db 'Complete Address:', 10
p_address_len equ $ - p_address

summary_heading db 10, '--- STUDENT INFORMATION SUMMARY ---', 10
summary_heading_len equ $ - summary_heading

l_status db 'Enrollment Status: '
l_status_len equ $ - l_status

l_student_no db 'Student Number: '
l_student_no_len equ $ - l_student_no

l_name db 'Name: '
l_name_len equ $ - l_name

l_phone db 'Phone: '
l_phone_len equ $ - l_phone

l_zip db 'ZIP Code: '
l_zip_len equ $ - l_zip
```

```
l_college db 'College: '
l_college_len equ $ - l_college

l_program db 'Program: '
l_program_len equ $ - l_program

l_year_level db 'Year Level: '
l_year_level_len equ $ - l_year_level

l_term db 'Term: '
l_term_len equ $ - l_term

l_school_year db 'School Year: '
l_school_year_len equ $ - l_school_year

l_address db 'Address: '
l_address_len equ $ - l_address

subjects_heading db 10, '--- SUBJECTS AND UNITS ---', 10
                 db 'Enter 0 units if you are not taking a subject.', 10
subjects_heading_len equ $ - subjects_heading

units_prompt db '   Units for this subject (whole number): ', 10
units_prompt_len equ $ - units_prompt

unit_summary_heading db 10, '--- SUBJECT UNIT SUMMARY ---', 10
unit_summary_heading_len equ $ - unit_summary_heading

units_label db '  Units: '
units_label_len equ $ - units_label

fee_heading db 10, '--- TUITION FEE SUMMARY ---', 10
fee_heading_len equ $ - fee_heading

label_total_units db 'Total units: '
label_total_units_len equ $ - label_total_units

label_unit_price db 'Unit price: PHP '
label_unit_price_len equ $ - label_unit_price

label_tuition db 'Tuition (total units x PHP 1,000): PHP '
label_tuition_len equ $ - label_tuition
```

```
    label_lab_subjects db 'Laboratory subjects taken: '
    label_lab_subjects_len equ $ - label_lab_subjects

    label_lab_fee db 'Laboratory fee (PHP 2,500 each): PHP '
    label_lab_fee_len equ $ - label_lab_fee

    label_total_due db 'TOTAL ESTIMATED TUITION + LAB FEES: PHP '
    label_total_due_len equ $ - label_total_due

    footer db 10, 'Note: This estimate excludes miscellaneous, ID validation,', 10
           db 'and any other fees not specified in the calculation.', 10
    footer_len equ $ - footer

    newline db 10
    newline_len equ $ - newline

    zero_text db '0', 0

    subj1  db '1. CCS0021  - INFORMATION MANAGEMENT (LEC)', 0
    subj2  db '2. CCS0021L - INFORMATION MANAGEMENT (LAB)', 0
    subj3  db '3. CS0001   - DISCRETE STRUCTURES I', 0
    subj4  db '4. CS0003   - COMPUTER SYSTEMS & ARCHITECTURE', 0
    subj5  db '5. CS0003L  - COMPUTER SYSTEMS & ARCHITECTURE (LAB)', 0
    subj6  db '6. CS0070   - OBJECT ORIENTED PROGRAMMING', 0
    subj7  db '7. CS0070L  - OBJECT ORIENTED PROGRAMMING-LAB', 0
    subj8  db '8. GED0021  - SPECIALIZED ENGLISH PROGRAM 2', 0
    subj9  db '9. GED0055  - MATHEMATICAL ANALYSIS', 0
    subj10 db '10. GED0081  - COLLEGE PHYSICS 1 LECTURE', 0
    subj11 db '11. GED0081L - COLLEGE PHYSICS 1 LABORATORY', 0

    subject_table dd subj1, subj2, subj3, subj4, subj5, subj6
                  dd subj7, subj8, subj9, subj10, subj11

    lab_flags db 0, 1, 0, 0, 1, 0, 1, 0, 0, 0, 1

    php_unit_price equ 1000
    php_lab_price equ 2500

SECTION .bss

    status_buf      resb 128
    student_no_buf  resb 128
    name_buf        resb 128
```

```
    phone_buf       resb 128
    zip_buf         resb 128
    college_buf     resb 128
    program_buf     resb 128
    year_level_buf  resb 128
    term_buf        resb 128
    school_year_buf resb 128
    address_buf     resb 128

    integer_buffer resb 32

    units_array resd 11

    total_units resd 1
    lab_subject_count resd 1
    tuition_amount resd 1
    lab_amount resd 1
    total_amount resd 1

    digit_buffer resb 16

SECTION .text

_start:

    PRINT heading, heading_len
    PRINT student_heading, student_heading_len

    ASK_STRING p_status, p_status_len, status_buf, 128
    ASK_STRING p_student_no, p_student_no_len, student_no_buf, 128
    ASK_STRING p_name, p_name_len, name_buf, 128
    ASK_STRING p_phone, p_phone_len, phone_buf, 128
    ASK_STRING p_zip, p_zip_len, zip_buf, 128
    ASK_STRING p_college, p_college_len, college_buf, 128
    ASK_STRING p_program, p_program_len, program_buf, 128
    ASK_STRING p_year_level, p_year_level_len, year_level_buf, 128
    ASK_STRING p_term, p_term_len, term_buf, 128
    ASK_STRING p_school_year, p_school_year_len, school_year_buf, 128
    ASK_STRING p_address, p_address_len, address_buf, 128

    PRINT summary_heading, summary_heading_len

    PRINT l_status, l_status_len
```

```
push dword status_buf
call print_string
add esp, 4
PRINT newline, newline_len

PRINT l_student_no, l_student_no_len
push dword student_no_buf
call print_string
add esp, 4
PRINT newline, newline_len

PRINT l_name, l_name_len
push dword name_buf
call print_string
add esp, 4
PRINT newline, newline_len

PRINT l_phone, l_phone_len
push dword phone_buf
call print_string
add esp, 4
PRINT newline, newline_len

PRINT l_zip, l_zip_len
push dword zip_buf
call print_string
add esp, 4
PRINT newline, newline_len

PRINT l_college, l_college_len
push dword college_buf
call print_string
add esp, 4
PRINT newline, newline_len

PRINT l_program, l_program_len
push dword program_buf
call print_string
add esp, 4
PRINT newline, newline_len

PRINT l_year_level, l_year_level_len
push dword year_level_buf
```

```
    call print_string
    add esp, 4
    PRINT newline, newline_len

    PRINT l_term, l_term_len
    push dword term_buf
    call print_string
    add esp, 4
    PRINT newline, newline_len

    PRINT l_school_year, l_school_year_len
    push dword school_year_buf
    call print_string
    add esp, 4
    PRINT newline, newline_len

    PRINT l_address, l_address_len
    push dword address_buf
    call print_string
    add esp, 4
    PRINT newline, newline_len

    mov dword [total_units], 0
    mov dword [lab_subject_count], 0

    PRINT subjects_heading, subjects_heading_len

    xor esi, esi

.collect_subjects:

    cmp esi, 11
    jge .subjects_done

    mov eax, [subject_table + esi*4]
    push eax
    call print_string
    add esp, 4

    PRINT newline, newline_len
    PRINT units_prompt, units_prompt_len

    call read_integer
```

```
    mov [units_array + esi*4], eax
    add [total_units], eax

    cmp eax, 0
    jle .next_subject

    cmp byte [lab_flags + esi], 1
    jne .next_subject

    inc dword [lab_subject_count]

.next_subject:

    inc esi
    jmp .collect_subjects

.subjects_done:

    mov eax, [total_units]
    imul eax, php_unit_price
    mov [tuition_amount], eax

    mov eax, [lab_subject_count]
    imul eax, php_lab_price
    mov [lab_amount], eax

    mov eax, [tuition_amount]
    add eax, [lab_amount]
    mov [total_amount], eax

    PRINT unit_summary_heading, unit_summary_heading_len

    xor esi, esi

.show_subjects:

    cmp esi, 11
    jge .show_fees

    mov eax, [subject_table + esi*4]
    push eax
    call print_string
```

```
    add esp, 4

    PRINT units_label, units_label_len

    push dword [units_array + esi*4]
    call print_uint
    add esp, 4

    PRINT newline, newline_len

    inc esi
    jmp .show_subjects

.show_fees:

    PRINT fee_heading, fee_heading_len

    PRINT label_total_units, label_total_units_len
    push dword [total_units]
    call print_uint
    add esp, 4
    PRINT newline, newline_len

    PRINT label_unit_price, label_unit_price_len
    push dword php_unit_price
    call print_uint
    add esp, 4
    PRINT newline, newline_len

    PRINT label_tuition, label_tuition_len
    push dword [tuition_amount]
    call print_uint
    add esp, 4
    PRINT newline, newline_len

    PRINT label_lab_subjects, label_lab_subjects_len
    push dword [lab_subject_count]
    call print_uint
    add esp, 4
    PRINT newline, newline_len

    PRINT label_lab_fee, label_lab_fee_len
    push dword [lab_amount]
```

```
    call print_uint
    add esp, 4
    PRINT newline, newline_len

    PRINT label_total_due, label_total_due_len
    push dword [total_amount]
    call print_uint
    add esp, 4
    PRINT newline, newline_len

    PRINT footer, footer_len

    mov eax, 1
    xor ebx, ebx
    int 0x80

print_string:

    push ebp
    mov ebp, esp

    push ebx
    push esi

    mov esi, [ebp+8]
    xor edx, edx

.count_chars:

    cmp byte [esi+edx], 0
    je .write_chars

    inc edx
    jmp .count_chars

.write_chars:

    test edx, edx
    jz .print_done

    mov eax, 4
    mov ebx, 1
    mov ecx, esi
```

```
    int 0x80

.print_done:

    pop esi
    pop ebx

    mov esp, ebp
    pop ebp
    ret

read_line:

    push ebp
    mov ebp, esp

    push ebx
    push esi
    push edi

    mov esi, [ebp+8]
    mov edx, [ebp+12]
    dec edx

    mov eax, 3
    xor ebx, ebx
    mov ecx, esi
    int 0x80

    test eax, eax
    jle .empty_line

    mov edi, eax
    mov byte [esi+edi], 0

    cmp byte [esi+edi-1], 10
    jne .check_cr

    mov byte [esi+edi-1], 0
    dec edi

.check_cr:
```

```
    cmp edi, 0
    je .line_done

    cmp byte [esi+edi-1], 13
    jne .line_done

    mov byte [esi+edi-1], 0
    jmp .line_done

.empty_line:

    mov byte [esi], 0

.line_done:

    pop edi
    pop esi
    pop ebx

    mov esp, ebp
    pop ebp
    ret

read_integer:

    push ebp
    mov ebp, esp

    push ebx
    push esi
    push edx

    push dword 32
    push dword integer_buffer
    call read_line
    add esp, 8

    mov esi, integer_buffer
    xor eax, eax

.parse_digit:

    mov bl, [esi]
```

```
    cmp bl, 0
    je .integer_done

    cmp bl, '0'
    jb .integer_done

    cmp bl, '9'
    ja .integer_done

    imul eax, eax, 10
    sub bl, '0'
    movzx edx, bl
    add eax, edx

    inc esi
    jmp .parse_digit

.integer_done:

    pop edx
    pop esi
    pop ebx

    mov esp, ebp
    pop ebp
    ret

print_uint:

    push ebp
    mov ebp, esp

    push ebx
    push esi
    push edi

    mov eax, [ebp+8]

    test eax, eax
    jnz .convert_number

    push dword zero_text
```

```
    call print_string
    add esp, 4
    jmp .number_done

.convert_number:

    mov edi, digit_buffer
    add edi, 15

    mov byte [edi], 0
    mov ebx, 10

.convert_loop:

    xor edx, edx
    div ebx

    add dl, '0'
    dec edi
    mov [edi], dl

    test eax, eax
    jnz .convert_loop

    push edi
    call print_string
    add esp, 4

.number_done:

    pop edi
    pop esi
    pop ebx

    mov esp, ebp
    pop ebp
    ret
```